

Jomon George Reji

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Professional Profile

Master's student in **Mechatronics and Cyber-Physical Systems** at **Technische Hochschule Deggendorf** with strong interests in **Robotics, Autonomous Systems, Artificial Intelligence, SLAM, Sensor Fusion, Computer Vision, Autonomous Driving, AR/VR, Embedded Systems, and Industrial Automation.**

Experienced in developing **ROS-based robotic systems, LiDAR-based localization and mapping solutions, simulation environments, and sensor-driven control systems** using **Python, C++, MATLAB, ROS/ROS2, Gazebo, Unity, and Simulink.**

Education

M.Sc. Mechatronics and Cyber-Physical Systems

Mar 2023 – Present

Technische Hochschule Deggendorf, Germany

Relevant Areas: **Robotics, Autonomous Systems, Machine Learning, Computer Vision, Sensor Fusion, Embedded Systems, Simulation**

Status: All coursework completed. Master's thesis pending.

B.Tech. Mechanical Engineering

Aug 2017 – Jun 2021

India

CGPA: **8.09 / 10**

Technical Skills

Programming: Python, C++, MATLAB, C#, SQL (Basic)

Robotics: ROS, ROS2, SLAM, Localization, Mapping, Navigation, Path Planning

Artificial Intelligence: Machine Learning, Deep Learning Fundamentals, PyTorch Basics, Computer Vision Fundamentals

Simulation: Gazebo, Unity, MATLAB/Simulink

Sensors & Perception: LiDAR, IMU, Camera Systems, Sensor Fusion Fundamentals

AR/VR: Unity XR, Virtual Reality Simulation Development

Tools: Linux, Git, Visual Studio

Projects

Inventory Robot with SLAM (ROS)

Technologies: ROS, Python, LiDAR, Gazebo

- Developed a **ROS-based Autonomous Navigation System** using **LiDAR-based SLAM.**
- Implemented **Localization, Mapping, Sensor Data Processing,** and Navigation workflows.
- Evaluated navigation performance in simulation environments.

VR Safety Training for Industrial Oven

Technologies: Unity, C#

- Developed immersive **Virtual Reality Training Environment.**
- Implemented industrial safety simulations and interaction systems.

Dual Axis Solar Tracking System

Technologies: MATLAB, Simulink

- Designed sensor-driven tracking system.

- Performed simulation-based optimization and control analysis.

Sustainable Razor Design using Additive Manufacturing

Technologies: CAD, Additive Manufacturing

- Designed and optimized product for additive manufacturing.
- Applied sustainable engineering principles.

Professional Experience

Amazon Germany

L1 Associate

Current

- Problem Solve Operations
- Ambassador Training
- Buffer Management
- Yard Marshal Support
- Excel-based Reporting and Process Monitoring

Adecco Germany

Warehouse & Logistics Associate

- Pick and Stage Operations
- Inventory Handling
- Logistics Support

Junior Engineer Trainee – Sharda Motors

Dec 2022 – Mar 2023

- Manufacturing support
- Quality inspection
- Testing and validation
- Engineering documentation

Graduate Mechanical Trainee – Newfriday Engineering

Jan 2022 – Dec 2022

- Mechanical engineering support
- Manufacturing workflows
- Engineering documentation

Training

Amazon Internal Training

- Problem Solve Training
- Ambassador Training
- Buffer Management
- Safety Procedures
- Yard Marshal Operations

Research Interests

- Autonomous Driving
 - Mobile Robotics
 - SLAM and Localization
 - Sensor Fusion
 - Computer Vision
 - AI for Robotics
 - AR/VR
 - Human-Robot Interaction
 - Planetary Robotics
 - Industrial Automation

Languages

- English (**Fluent**)
- German (**A2**)
- Malayalam (**Native**)
- Hindi (**Conversational**)